

Learned Map-Completion Priors for Mobile-Robot Frontier Exploration

Final Project Proposal (FP2)

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Format: Solo · ROS 2 Humble · Stage simulator · builds directly on PA3 + PA4.

Track: Frontier Explorer (per the spec’s example tracks).

Integrity & AI use. This work is my own. AI helped me with LaTeX formatting and literature search. Collaborators: none. Signature: Moin Mattar.

Abstract

Frontier-based exploration scores each candidate frontier with a hand-crafted heuristic that is blind to the structure of the unseen environment. We propose to learn a generative prior over completed occupancy grids using a small **conditional denoising diffusion model**, and use samples from that prior to score frontiers by *expected* information gain with an *uncertainty bonus*. The method is trained on synthetic (partial-map, full-map) pairs from the **HouseExpo** dataset and evaluated in Stage on the maze world from PA3 plus additional procedurally generated worlds. We hypothesize that sampling-based scoring reduces time-to-coverage relative to standard frontier exploration, with the largest gains in environments with structural ambiguity (corridor vs. dead-end). We discuss extensions to **Diffusion Forcing** for variable-horizon rollouts as future work.

1. Introduction

Standard frontier-based exploration (Yamauchi 1997; Quin et al. 2014, the algorithm covered in Lec 19) picks the next frontier with a hand-crafted score, typically:

$$\text{score}(f) = \alpha \cdot N_{\text{unknown}}(f) - \beta \cdot d(r, f),$$

where $N_{\text{unknown}}(f)$ is the number of unknown cells visible from frontier f and $d(r, f)$ is the robot’s distance to it. This heuristic ignores **what lies beyond the frontier**: a frontier opening into a wide unmapped room and a frontier opening into a dead-end closet can score identically; only after the robot drives there does the difference manifest.

Learned map-completion priors address this. Shrestha et al. (ICRA 2019) trained a conditional GAN to predict the full map from a partial observation and used the predicted free-space to inform frontier scoring. More recent work uses diffusion models (Lin et al. 2024, arXiv:2409.10681) for 3D occupancy prediction at the frontier. Both these methods produce a **single deterministic prediction** and use it as a point estimate of “what’s behind the wall.”

Our contribution is to treat the unseen environment as **inherently uncertain** and score frontiers by the *distribution* of plausible completions. For each candidate frontier f , we sample K completions from the diffusion model, then score f by $\mathbb{E}[\text{info gain}] + \lambda \cdot \text{Var}[\text{info gain}]$. The variance bonus encourages the robot toward frontiers where the model is *uncertain*, which are precisely the frontiers where new observations are most informative.

2. Method

2.1 Map Completion Model

Architecture. A small conditional U-Net ($\sim 5\text{M}$ parameters) operating on 256×256 occupancy grid patches. Inputs: a partial grid where each cell is one of {free, occupied, unknown} (encoded as a 3-channel one-hot image). Output: a predicted full grid (3 channels, softmaxed per cell).

Training objective. Standard conditional DDPM (Ho et al. 2020):

$$\mathcal{L} = \mathbb{E}_{t, x_0, \epsilon} [\|\epsilon - \epsilon_\theta(x_t, t, c)\|^2],$$

where c is the partial map (conditioning) and x_0 is the ground-truth full map.

Training data. **HouseExpo** dataset (Li et al. IROS 2020): 35,000+ indoor floor plans. For each map we generate (partial, full) pairs by simulating a partial-observation mask: place a virtual robot at a random cell, cast a 360° lidar with realistic range and angular resolution (matching the Stage rosbots lidar), mark observed cells as known, the rest as unknown. The resulting partial map is the conditioning c , the original full map is x_0 . This yields $\sim 500\text{k}$ training samples cheaply.

2.2 Sampling-Based Frontier Scoring

For each candidate frontier f at inference time:

1. Crop the current partial map to a 256×256 window centered on the robot.
2. Sample $K = 8$ completions from the diffusion model conditioned on the partial map.
3. For each completion, compute the realized info gain: the number of currently-unknown cells that the completion predicts as **free and reachable from** f (within a configurable radius).
4. Score f by

$$\text{score}(f) = \mathbb{E}_k[g_k(f)] + \lambda \cdot \text{Std}_k[g_k(f)] - \beta \cdot d(r, f),$$

where $g_k(f)$ is the info gain under completion k , λ controls the uncertainty bonus, and the last term is the standard distance penalty.

5. Pick $f^* = \arg \max_f \text{score}(f)$, drive to it, scan, update the partial map, repeat.

2.3 ROS 2 Integration

The system is one ROS 2 node that subscribes to `/map` (from a slightly modified PA4 mapper that also tracks frontier cells) and publishes both `/predicted_map` (the mean completion, for visualization) and `/scored_frontiers` (a `geometry_msgs/PoseArray`, each pose annotated with its score). A simple pure-pursuit controller drives the robot to the highest-scoring frontier; the loop continues.

3. Connection to Course Material

Class component	Where in this project
Lec 10, occupancy grids	The map representation; <code>pa4.grid_mapper.py</code> is the substrate.
Lec 12, search-based planning	A* used to compute $d(r, f)$ and reachability inside each completion.
Lec 14–16, Bayes filtering	The PA4 log-odds update; the diffusion model is a learned generative version of the same posterior.
Lec 19, exploration & coverage	The frontier-based exploration loop (Quin et al. 2014).
Lec 20, ML for robotics	The conditional diffusion model is the ML component.

The project synthesizes **five distinct lecture units** into one autonomous system.

4. Experimental Plan

4.1 Worlds

PA3 maze (small, structured), two procedurally generated mazes (one with dead-ends, one with one large central room), and two HouseExpo floorplans loaded into Stage (out-of-training-distribution). **Five worlds $\times$ ten trials each = 50 trials per condition.**

4.2 Baselines

- **Heuristic frontier.** Standard $\text{score}(f) = N_{\text{unknown}}(f) - \beta d(r, f)$.
- **Shrestha-style.** A deterministic conditional U-Net (no diffusion) trained on the same data, single prediction used for info gain. Isolates “stochastic sampling” from “learned prior.”
- **Ours (no variance bonus).** $\lambda = 0$. Isolates “expected info gain” from “uncertainty-seeking.”
- **Ours (full).** $\lambda > 0$.

4.3 Metrics

- **Time-to-90%-coverage** (primary).
- **Path length** to 90% coverage.
- **Info-gain calibration:** reliability diagram of predicted vs. realized info gain.
- **Qualitative:** side-by-side visualization of $K = 8$ completions per frontier (the headline demo figure).

4.4 Ablations

$K \in \{1, 4, 8, 16\}$ (sample-count sweep); $\lambda \in \{0, 0.5, 1.0, 2.0\}$ (variance-bonus sweep); training-data scale 10%/50%/100%.

5. Risks and Fallbacks

Risk	Mitigation
Diffusion training won't converge in time.	Diffusion U-Nets are well-trodden territory. Many working open-source implementations exist (huggingface/diffusers , lucidrains/denoising-diffusion-pytorch). If training fails, swap for a deterministic U-Net regressor: degrades to the Shrestha-style baseline but still novel (sampling becomes dropout-based).
Sim-to-sim domain gap (HouseExpo $\rightarrow$ Stage).	Include 2 Stage worlds in training. Domain-randomize lidar noise.
ROS 2 integration eats time.	The mapper, frontier extractor, and pure-pursuit controller are already implemented in PA3/PA4. Only the diffusion-inference module is new.
Inference too slow for real-time.	At 256×256 and $K = 8$, denoising takes $\sim 1\text{--}2\text{s}$ on a laptop GPU. Acceptable because frontier selection happens once every ~ 5 seconds during driving.

6. Future Work, Diffusion Forcing for Variable-Horizon Rollouts

The current method predicts a *single-shot* completion. A natural extension is to predict the **evolution of the map over time** as the robot moves toward a candidate frontier, a video-generation problem rather than image-completion. **Diffusion Forcing** (Chen et al., NeurIPS 2024) is purpose-built for this: it trains a sequence model with independent per-token noise levels, enabling variable-horizon rollouts with calibrated per-step uncertainty. If time permits past FP3 (June 1), I will adapt the open-source [buoyancy99/diffusion-forcing](#) repo to roll out (map_patch, action) sequences and score frontiers by expected info gain over a 20-step horizon. **This is flagged as stretch work, not a deliverable.**

7. Timeline

Date	Milestone
May 20–22	Repo setup. Reproduce a HuggingFace diffusion tutorial on MNIST to confirm pipeline.
May 22–26	Data generation: HouseExpo loader, partial-map simulator, $\sim 500\text{k}$ training samples cached.
May 26–27	First training run begins. FP2.5 in-class progress update (May 27) : show loss curves + sample completions.
May 27–30	Training converges. Implement frontier scorer. Integrate into PA4 ROS node.
May 30–Jun 1	Run baselines + ablations in Stage. Generate plots. FP3 (June 1) : 6-min presentation + demo video.
Jun 1–Jun 6	Write 4-page IEEE paper. Polish demo. Polish github repo. FP4 (June 6) : final report + code + 3-min teaser.

8. Deliverables Summary

4-page IEEE-format final report; open-source github repo with training + inference + ROS 2 integration; demo video showing the system in three Stage worlds; quantitative comparison against 3 baselines on 5 worlds $\times$ 10 trials.

9. References

- Yamauchi, B. *A frontier-based approach for autonomous exploration*. CIRA 1997.
- Quin, P. et al. *Frontier-based multi-robot exploration with rendezvous*. 2014 (Lec 19).
- Shrestha, R. et al. *Learned map prediction for enhanced mobile robot exploration*. ICRA 2019.
- Ho, J., Jain, A., Abbeel, P. *Denoising diffusion probabilistic models*. NeurIPS 2020.
- Li, T. et al. *HouseExpo: a large-scale 2D indoor layout dataset for learning-based algorithms on mobile robots*. IROS 2020.
- Lin et al. *Online diffusion-based 3D occupancy prediction at the frontier with probabilistic map reconciliation*. 2024. arXiv:2409.10681.
- Chen, B. et al. *Diffusion forcing: next-token prediction meets full-sequence diffusion*. NeurIPS 2024. arXiv:2407.01392.

Talking points for the meeting (last page of the doc):

- **One sentence:** learn a generative prior over completed occupancy grids; sample K completions; score frontiers by expected info gain plus an uncertainty bonus.
- **Why not Diffusion Forcing as the main deliverable:** adds complexity I can't safely deliver in three weeks solo. Single-shot version is cleaner to scope and evaluate; DF flagged as future work.
- **Novelty over Shrestha 2019 / Lin 2024:** they predict one deterministic map and treat it as truth. We sample, average, and add a variance bonus, so the robot is aware of its own uncertainty.
- **Ask the prof:** (1) does the novelty claim land; (2) Stage vs. Gazebo; (3) real robot at the end or sim-only; (4) is "DF as future work" the right framing.
- **Insurance:** if diffusion training fails, the model degrades cleanly to a deterministic regressor. Sampling-based scoring still works (via dropout). Guaranteed minimum deliverable.